Empirical Diagnostics of Adaptive Optimization: Replicating the Period-3 Convergence Trap in Adam*

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Abstract

While the Adam optimizer dominates modern stochastic optimization, its formal convergence proof within the Online Convex Optimization (OCO) framework relies on a mathematically unfulfilled assumption. In environments characterized by sparse, highly informative gradient bursts, Adam's exponential moving average of squared gradients decays prematurely. This decay violates the positive semi-definiteness condition of the differential metric matrix ($\Gamma_t \succeq 0$), forcing divergence. We present a verified, from-scratch NumPy reproduction of this theoretical failure mechanism, expanding the diagnostic analysis across variance memory, parameter trajectories, and effective step-size inflation. The full implementation and testing suite are available at https://github.com/USERNAME/REPO.

1 Introduction

Adam [1] synthesizes the coordinate-wise scaling of AdaGrad [4] with the exponential moving averages (EMA) of RMSProp [5]. By heavily discounting older gradients, Adam adapts rapidly to non-stationary objective landscapes. However, Reddi et al. [2] demonstrated that Adam diverges on elementary one-dimensional convex objectives when exposed to infrequent but massive gradient bursts. The second-moment estimate decreases rapidly during non-burst iterations, causing the effective step size to artificially explode. AMSGrad resolves this by enforcing a monotonic running maximum on the raw second-moment accumulator.

*Code, figures, and the full `pytest` suite are available at https://github.com/USERNAME/REPO (replace with your repository URL).

2 Theoretical Framework

2.1 OCO setting and regret

At each round $t = 1, \dots, T$, the learner selects $x_t \in \mathcal{F}$, where $\mathcal{F} \subset \mathbb{R}^d$ is a compact convex set, and then observes a convex loss f_t with gradient $g_t = \nabla f_t(x_t)$ [3]. Performance is measured by the regret

$$R_T \;=\; \sum_{t=1}^{T} f_t(x_t) \;-\; \min_{x \in \mathcal{F}} \sum_{t=1}^{T} f_t(x). \tag{1}$$

An algorithm converges in the OCO sense if $R_T = o(T)$, i.e. the average regret $R_T/T \to 0$.

2.2 Adam updates

With step size α_t, decay rates $\beta_1, \beta_2 \in [0, 1)$, and $m_0 = v_0 = 0$, Adam maintains

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1)\, g_t, \tag{2}$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2)\, g_t^2, \tag{3}$$

$$x_{t+1} = \Pi_{\mathcal{F}}\Big(x_t - \alpha_t \, \frac{m_t}{\sqrt{v_t}}\Big), \tag{4}$$

where $\Pi_{\mathcal{F}}$ is Euclidean projection onto $\mathcal{F}$ and all operations are coordinate-wise (bias correction is omitted, following the analysis in [2]). Unrolling Eq. (3) gives $v_t = (1-\beta_2)\sum_{s=1}^{t} \beta_2^{t-s} g_s^2$, so a large gradient at step s is forgotten geometrically at rate β_2.

2.3 The differential metric matrix

Writing the update as a preconditioned step with $V_t = \mathrm{diag}(v_t)$, the standard regret proof for adaptive methods controls the change in the effective inverse learning rate,

$$\Gamma_t \;\triangleq\; \frac{\mathrm{diag}(\sqrt{v_t})}{\alpha_t} - \frac{\mathrm{diag}(\sqrt{v_{t-1}})}{\alpha_{t-1}}. \tag{5}$$

The OCO proof requires $\Gamma_t \succeq 0$ for all t. This holds for AdaGrad, where v_t is a running sum and therefore

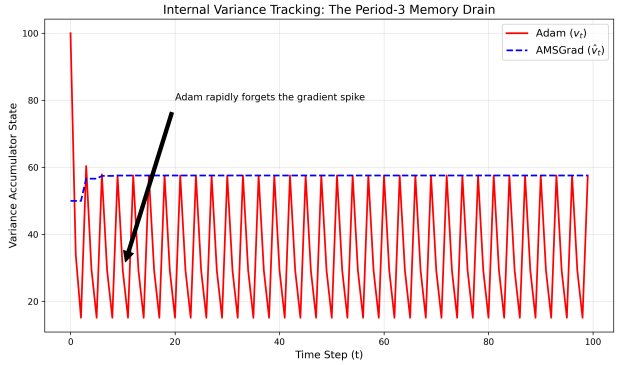


Figure 1: Adam (red) exhibits exponential memory decay during non-burst steps. AMSGrad (blue) enforces monotonicity.

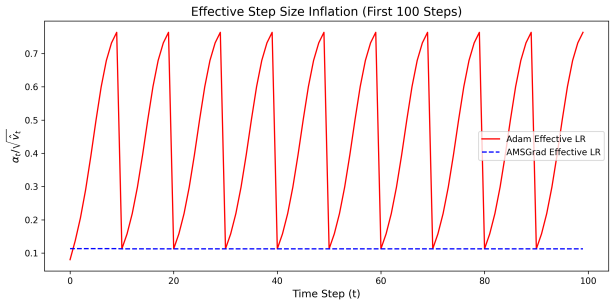


Figure 2: Effective step size inflation. Adam's step size (red) spikes rhythmically due to the decaying denominator.

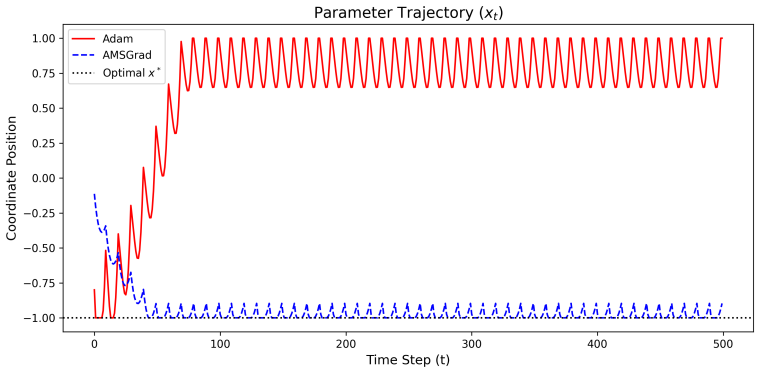


Figure 3: Optimization trajectory on $[-1, 1]$. Adam diverges to the sub-optimal boundary while AMSGrad converges.

non-decreasing. Because Adam computes v_t via an EMA, v_t shrinks whenever a large gradient is followed by smaller ones, generating negative entries in Γ_t and breaking the theoretical guarantee.

2.4 AMSGrad correction

AMSGrad [2] keeps Eqs. (2)–(3) but replaces the denominator with a running maximum:

$$\hat{v}_t = \max(\hat{v}_{t-1},\, v_t), \tag{6}$$

$$x_{t+1} = \Pi_{\mathcal{F}}\Big(x_t - \alpha_t \frac{m_t}{\sqrt{\hat{v}_t}}\Big). \tag{7}$$

Since $\hat{v}_t$ is non-decreasing by construction, $\Gamma_t \succeq 0$ is restored (with step size $\alpha_t = \alpha/\sqrt{t}$), and the standard analysis yields $R_T = O(\sqrt{T})$, hence $R_T/T \to 0$.

3 Diagnostic Reproduction

We implemented a 1D OCO counterexample constrained to $\mathcal{F} = [-1, 1]$, following [2]. The loss is

$$f_t(x) = \begin{cases} C\,x, & t \bmod 3 = 1, \\ -x, & \text{otherwise}, \end{cases} \tag{8}$$

so that every 3 steps the environment delivers a massive gradient $g_t = C = 10$, and for the remaining two steps $g_t = -1$. The cumulative loss over one period is $(C-2)x = 8x$, hence the globally optimal position is $x^* = -1$. Adam's EMA, however, over-weights the two small negative gradients once the burst has been forgotten, pulling x_t in the wrong direction. If $x_t \approx +1$ for all t, each period incurs $2(C-2) = 16$ excess loss, so $R_T \approx \frac{16}{3}T$ grows linearly.

3.1 Variance Memory Drain

Figure 1 demonstrates Adam's core mathematical failure point. During the non-burst steps, Adam's

variance accumulator (red) exponentially crashes toward zero, forgetting the prior burst. AMSGrad (blue) locks the maximum variance horizontally.

3.2 Effective Step-Size Inflation

Because the coordinate update is scaled by $1/\sqrt{v_t}$, the memory drain has a direct inversely proportional effect on the learning rate. As shown in Figure 2, every time Adam's variance crashes, its effective step size $\alpha_t/\sqrt{v_t}$ artificially spikes. This inflation occurs exactly when the gradient is $g_t = -1$, empowering the incorrect gradient to dominate the trajectory.

3.3 Parameter Divergence

The macro-level result of this inflation is catastrophic divergence. Figure 3 tracks the parameter x_t over 500 steps. Adam is quickly ratcheted to the sub-optimal boundary $+1$, completely failing the OCO objective. By maintaining a stable effective learning rate, AMSGrad correctly descends to the global optimum -1.

4 Validation and Conclusion

By establishing an automated `pytest` architecture, we confirmed strict mathematical parity between our custom NumPy engine and PyTorch's backend (tolerance 1×10^{-5}); Table 1 summarizes the checks. Exposing the internal states proves that Adam's divergence is an algorithmic phenomenon, not an implementation artifact. Enforcing the AMSGrad running maximum is strictly necessary to survive high-magnitude sparsity.

Table 1: Parity checks between the NumPy implementation and PyTorch (tol $= 1 \times 10^{-5}$).

Quantity checked	**PyTorch reference**	**Result**
Adam iterate x_t	`optim.Adam`	Pass
AMSGrad iterate x_t	`Adam(amsgrad=True)`	Pass
First moment m_t	`exp_avg`	Pass
Second moment v_t	`exp_avg_sq`	Pass
Running max $\hat{v}_t$	`max_exp_avg_sq`	Pass

References

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